

mambavision的环境安装

1、安装环境

(1) 创建环境

```
conda create -n mambavision python==3.10.3
```

(2) 安装torch、torchvision

```
conda activate mambavision pip install torch==2.1.0 torchvision==0.16.0 torchaudio==2.1.0 --index-url https://download.pytorch.org/whl/cu121
```

(3) 安装mmdcv

```
pip install mmdcv==2.1.0 -f https://download.openmmlab.com/mmdcv/dist/cu121/torch2.1/index.html
```

我这里之前装了mmseg的环境，就直接复制之前创建的mmseg的环境版本

```
(base) adam@adam-computer:~$ conda create --name mambavision --clone mmseg_env
2 channel Terms of Service accepted
Retrieving notices: done
Source: /mnt/d/App_path/miniconda3/envs/mmseg_env
Destination: /mnt/d/App_path/miniconda3/envs/mambavision
Packages: 30
Files: 32313
```

(4) 安装ssm、causal

这里安装的版本是：causal_conv1d-1.5.0.post5 + mamba_ssm-2.2.3.post2

链接：

[Releases · Dao-AI Lab/causal-conv1d](https://github.com/Dao-AI Lab/causal-conv1d/releases?page=2)

[Releases · Dao-AI Lab/causal-conv1d](https://github.com/Dao-AI Lab/causal-conv1d/releases?page=2)

```
pip install mamba_ssm-2.2.3.post2+cu12torch2.1cxx11abiFALSE-cp310-cp310-linux_x86_64.whl
```

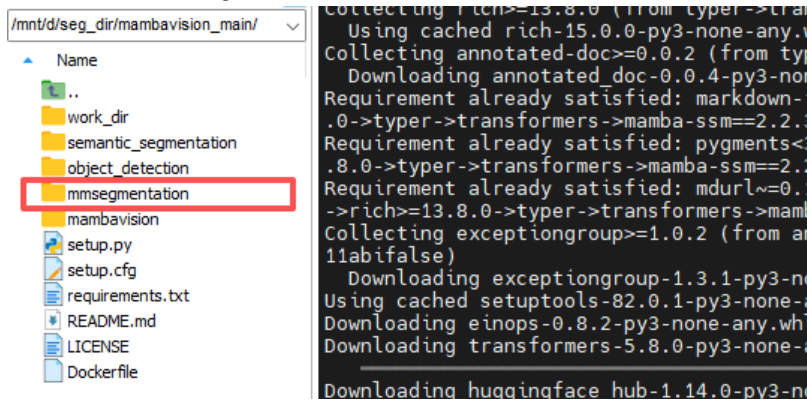
```
pip install causal_conv1d-1.5.0.post5+cu12torch2.1cxx11abiFALSE-cp310-cp310-linux_x86_64.whl
```

```
(mambavision) adam@adam-computer:/mnt/d/seg_dir/zip$ pip install causal_conv1d-1.5.0.post5+cu12torch2.1cxx11abiFALSE-cp310-cp310-linux_x86_64.whl
Processing ./causal_conv1d-1.5.0.post5+cu12torch2.1cxx11abiFALSE-cp310-cp310-linux_x86_64.whl
Requirement already satisfied: torch in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (2.1.0+cu121)
Requirement already satisfied: packaging in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (24.2)
Collecting ninja (from causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE)
  Downloading ninja-1.13.0-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl.metadata (5.1 kB)
Requirement already satisfied: filelock in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (3.14.0)
Requirement already satisfied: typing-extensions in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (4.15.0)
Requirement already satisfied: sympy in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (1.14.0)
Requirement already satisfied: networkx in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (3.4.2)
Requirement already satisfied: Jinja2 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (3.1.6)
Requirement already satisfied: fsspec in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (2026.2.0)
Requirement already satisfied: triton==2.1.0 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (2.1.0)
Requirement already satisfied: MarkupSafe>=2.0 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from Jinja2->torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (3.0.3)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from sympy->torch->causal_conv1d==1.5.0.post5+cu12torch2.1cxx11abiFALSE) (1.3.0)
Downloading ninja-1.13.0-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (180 kB)
Installing collected packages: ninja, causal_conv1d
Successfully installed causal_conv1d-1.5.0.post5 ninja-1.13.0
(mambavision) adam@adam-computer:/mnt/d/seg_dir/zip$ pip install mamba_ssm-2.2.3.post2+cu12torch2.1cxx11abiFALSE-cp310-cp310-linux_x86_64.whl
Processing ./mamba_ssm-2.2.3.post2+cu12torch2.1cxx11abiFALSE-cp310-cp310-linux_x86_64.whl
Requirement already satisfied: torch in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from mamba_ssm==2.2.3.post2+cu12torch2.1cxx11abiFALSE) (2.1.0+cu121)
Requirement already satisfied: ninja in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from mamba_ssm==2.2.3.post2+cu12torch2.1cxx11abiFALSE) (1.13.0)
Collecting einops (from mamba_ssm==2.2.3.post2+cu12torch2.1cxx11abiFALSE)
```

按提示进行版本的降低

```
Successfully uninstalled rich-13.4.2
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
openxlab 0.1.3 requires rich<=13.4.2, but you have rich 15.0.0 which is incompatible.
openxlab 0.1.3 requires setuptools<=60.2.0, but you have setuptools 82.0.1 which is incompatible.
Successfully installed annotated-doc-0.0.4 anyio-4.13.0 einops-0.8.2 exceptiongroup-1.3.1 h11-0.16.0 hf-xet-1.5.0 httpcore-1.0.9 httpx-0.28.1 huggingface-hub-1.14.0 mamba-ssm-2.2.3.post2 regex-2026.5.9 rich-15.0.0 safetensors-0.7.0 setuptools-82.0.1 shellingham-1.5.4 tokenizers-0.22.2 transformers-5.8.0 typer-0.25.1
(mambavision) adam@adam-computer: /mnt/d/seg_dir/zip$
```

(5) 下载了mmsegmentation的后，在通过文件安装依赖，方便后续训练自己数据的时候进行参数和文件配置的修改



```
Collecting rich<=13.8.0 (from typer->transformers)
Using cached rich-15.0.0-py3-none-any.whl (235 kB)
Collecting annotated-doc<=0.0.2 (from typer->transformers)
Downloading annotated_doc-0.0.4-py3-none-any.whl (11 kB)
Requirement already satisfied: markdown-it-py<3.0.0 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from typer->transformers->mamba-ssm==2.2.3.post2)
Requirement already satisfied: pygments<3.0.0 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from typer->transformers->mamba-ssm==2.2.3.post2)
Requirement already satisfied: mdurl<=0.1.2 in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from typer->rich==13.8.0->typer->transformers->mamba-ssm==2.2.3.post2)
Collecting exceptiongroup<=1.0.2 (from typer->transformers->mamba-ssm==2.2.3.post2)
Downloading exceptiongroup-1.3.1-py3-none-any.whl (18 kB)
Using cached setuptools-82.0.1-py3-none-any.whl (1.2 MB)
Downloading einops-0.8.2-py3-none-any.whl (16 kB)
Downloading transformers-5.8.0-py3-none-any.whl (4.0 MB)
Downloading huggingface-hub-1.14.0-py3-none-any.whl (415 kB)
```

cd mmsegmentation

pip install -v -e .

```
adding 'mmsegmentation-1.2.2.dist-info/top_level.txt'
adding 'mmsegmentation-1.2.2.dist-info/RECORD'
/tmp/pip-build-env-9t7f9cjb/overlay/lib/python3.10/site-packages/setuptools/command/editable_wheel.py:352: InformationOnly: Editable installation.
!!

*****
Please be careful with folders in your working directory with the same
name as your package as they may take precedence during imports.
*****

!!
with strategy, WheelFile(wheel_path, "w") as wheel_obj:
Building editable for mmsegmentation (pyproject.toml) ... done
Created wheel for mmsegmentation: filename=mmsegmentation-1.2.2-0.editable-py3-none-any.whl size=13774 sha256=a6fad2e49e18993b745580509fad99463ab5e147fb4c1b66b046702e866d576d
Stored in directory: /tmp/pip-ephem-wheel-cache-7g4d0mfx/wheels/82/5e/5f/e8596bc5f6b31d712101f50b09658e05edf9ef42aa1bcc35cf
Successfully built mmsegmentation
Installing collected packages: mmsegmentation
Successfully installed mmsegmentation-1.2.2
(mambavision) adam@adam-computer: /mnt/d/seg_dir/mambavision_main/mmsegmentation$
```

(6) 安装requirements.txt中指定版本的transformer

pip install transformers==4.50.0

(7) 安装指定版本的numpy和opencv防止冲突

\$pip install opencv-python==4.9.0.80\$

\$pip install numpy==1.26.4\$

(8) 按提示安装依赖包timm、ftfy

```
File "/mnt/d/seg_dir/mambavision_main/semantic_segmentation/tools/mamba_vision.py", line 14, in <module>
    from timm.models.registry import register_model
ModuleNotFoundError: No module named 'timm'
❖ (mambavision) adam@adam-computer: /mnt/d/seg_dir/mambavision_main$ pip install timm

Collecting timm
  Downloading timm-1.0.27-py3-none-any.whl.metadata (40 kB)
Requirement already satisfied: torch in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from timm) (2.1.0+cu121)
  Downloading timm-1.0.27-py3-none-any.whl (4.0 MB)
Installing collected packages: timm
Successfully installed timm-1.0.27

File "/mnt/d/seg_dir/mambavision_main/mmsegmentation/mmseg/utils/tokenizer.py", line 13, in <module>
    import ftfy
ModuleNotFoundError: No module named 'ftfy'
❖ (mambavision) adam@adam-computer: /mnt/d/seg_dir/mambavision_main$ pip install ftfy

Collecting ftfy
  Downloading ftfy-6.3.1-py3-none-any.whl.metadata (7.3 kB)
Requirement already satisfied: wcwidth in /mnt/d/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages (from ftfy) (0.7.0)
  Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)
Installing collected packages: ftfy
Successfully installed ftfy-6.3.1
```

(9) 遇到“ModuleNotFoundError: No module named 'pkg_resources'”

```
(mambavision) adam@adam-computer:/mnt/c/seg_dir/mambavision_main$ CUDA_VISIBLE_DEVICES=1 python semantic_segmentation/tools/train.py --config semantic_segmentation
nd_mamba_vision_160k_ade20k-512x512_small_our_method.py --work-dir ./work_dir/farmland_mamba_small_our_method --amp
warnings.warn(f"Importing from {__name__} is deprecated, please import via timm.models", FutureWarning)
/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/timm/models/layers/__init__.py:49: FutureWarning: Importing from timm.models.layers is dep
imm.layers
warnings.warn(f"Importing from {__name__} is deprecated, please import via timm.layers", FutureWarning)
05/23 10:37:16 - mmengine - WARNING - AMP training is already enabled in your config.
Traceback (most recent call last):
  File "/mnt/c/seg_dir/mambavision_main/semantic_segmentation/tools/train.py", line 116, in <module>
    main()
  File "/mnt/c/seg_dir/mambavision_main/semantic_segmentation/tools/train.py", line 104, in main
    runner = Runner.from_cfg(cfg)
  File "/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/mmengine/runner/runner.py", line 462, in from_cfg
    runner = cls(
  File "/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/mmengine/runner/runner.py", line 403, in __init__
    self._log_env(env_cfg)
  File "/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/mmengine/runner/runner.py", line 2368, in _log_env
    env = collect_env()
  File "/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/mmengine/utils/dl_utils/collect_env.py", line 73, in collect_env
    CUDA_HOME = _get_cuda_home()
  File "/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/mmengine/utils/dl_utils/collect_env.py", line 25, in _get_cuda_home
    from torch.utils.cpp_extension import CUDA_HOME
  File "/mnt/c/App_path/miniconda3/envs/mambavision/lib/python3.10/site-packages/torch/utils/cpp_extension.py", line 28, in <module>
    from pkg_resources import packaging # type: ignore[attr-defined]
ModuleNotFoundError: No module named 'pkg_resources'
```

```
pip install setuptools==60.0.0
```

```
● (mambavision) adam@adam-computer:/mnt/c/seg_dir/mambavision_main$ pip install setuptools==61.0.0
Collecting setuptools==61.0.0
  Downloading setuptools-61.0.0-py3-none-any.whl.metadata (6.0 kB)
  Downloading setuptools-61.0.0-py3-none-any.whl (1.1 MB)
    1.1/1.1 MB 4.3 MB/s 0:00:00
Installing collected packages: setuptools
  Attempting uninstall: setuptools
    Found existing installation: setuptools 60.0.0
    Uninstalling setuptools-60.0.0:
      Successfully uninstalled setuptools-60.0.0
  Successfully installed setuptools-61.0.0
❖ (mambavision) adam@adam-computer:/mnt/c/seg_dir/mambavision_main$
```

(10) 开始训练自己的数据任务

```
(mambavision) adam@adam-computer:/mnt/d/seg_dir/mambavision_main$ /mnt/d/App_path/miniconda3/envs/mambavision/bin/python /mnt/d/seg_dir/mambavision_main/se
mantic_segmentation/tools/train.py
05/11 16:00:35 - mmengine - INFO - paramwise_options -- backbone.outnorm3.weight:lr=5e-05
05/11 16:00:35 - mmengine - INFO - paramwise_options -- backbone.outnorm3.weight:weight_decay=0.0
05/11 16:00:35 - mmengine - INFO - paramwise_options -- backbone.outnorm3.weight:decay_mult=0.0
05/11 16:00:35 - mmengine - INFO - paramwise_options -- backbone.outnorm3.bias:lr=5e-05
05/11 16:00:35 - mmengine - INFO - paramwise_options -- backbone.outnorm3.bias:weight_decay=0.0
05/11 16:00:35 - mmengine - INFO - paramwise_options -- backbone.outnorm3.bias:decay_mult=0.0
05/11 16:00:35 - mmengine - WARNING - The prefix is not set in metric class IoUMetric.
05/11 16:00:36 - mmengine - WARNING - "FileClient" will be deprecated in future. Please use io functions in https://mmengine.readthedocs.io/en/latest/api/f
ileio.html#file-io
05/11 16:00:36 - mmengine - WARNING - "HardDiskBackend" is the alias of "LocalBackend" and the former will be deprecated in future.
05/11 16:00:36 - mmengine - INFO - Checkpoints will be saved to /mnt/d/seg_dir/mambavision_main/semantic_segmentation/work_dir.
05/11 16:00:47 - mmengine - INFO - Iter(train) [ 50/160000] base_lr: 1.6345e-06 lr: 1.6345e-06 eta: 10:15:31 time: 0.1722 data_time: 0.0093 memory:
16356 loss: 0.7930 decode.loss_ce: 0.5532 decode.acc_seg: 73.0653 aux.loss_ce: 0.2398 aux.acc_seg: 30.2965
05/11 16:00:56 - mmengine - INFO - Iter(train) [ 100/160000] base_lr: 3.3022e-06 lr: 3.3022e-06 eta: 8:57:18 time: 0.1713 data_time: 0.0090 memory:
4819 loss: 0.7513 decode.loss_ce: 0.5169 decode.acc_seg: 49.8800 aux.loss_ce: 0.2344 aux.acc_seg: 34.6289
05/11 16:01:05 - mmengine - INFO - Iter(train) [ 150/160000] base_lr: 4.9700e-06 lr: 4.9700e-06 eta: 8:32:04 time: 0.1727 data_time: 0.0089 memory:
4819 loss: 0.8106 decode.loss_ce: 0.5574 decode.acc_seg: 55.2383 aux.loss_ce: 0.2533 aux.acc_seg: 25.1947
05/11 16:01:13 - mmengine - INFO - Iter(train) [ 200/160000] base_lr: 6.6378e-06 lr: 6.6378e-06 eta: 8:19:18 time: 0.1743 data_time: 0.0093 memory:
4819 loss: 0.7521 decode.loss_ce: 0.5170 decode.acc_seg: 72.5470 aux.loss_ce: 0.2351 aux.acc_seg: 40.6200
05/11 16:01:22 - mmengine - INFO - Iter(train) [ 250/160000] base_lr: 8.3056e-06 lr: 8.3056e-06 eta: 8:11:20 time: 0.1701 data_time: 0.0089 memory:
4819 loss: 0.6655 decode.loss_ce: 0.4530 decode.acc_seg: 75.6109 aux.loss_ce: 0.2125 aux.acc_seg: 68.4864
05/11 16:01:31 - mmengine - INFO - Iter(train) [ 300/160000] base_lr: 9.9734e-06 lr: 9.9734e-06 eta: 8:06:24 time: 0.1754 data_time: 0.0099 memory:
4819 loss: 0.5721 decode.loss_ce: 0.3830 decode.acc_seg: 97.2919 aux.loss_ce: 0.1891 aux.acc_seg: 90.7919
05/11 16:01:39 - mmengine - INFO - Iter(train) [ 350/160000] base_lr: 1.1641e-05 lr: 1.1641e-05 eta: 8:02:52 time: 0.1740 data_time: 0.0094 memory:
4819 loss: 0.6755 decode.loss_ce: 0.4600 decode.acc_seg: 99.7744 aux.loss_ce: 0.2155 aux.acc_seg: 99.7744
05/11 16:01:48 - mmengine - INFO - Iter(train) [ 400/160000] base_lr: 1.3309e-05 lr: 1.3309e-05 eta: 8:00:28 time: 0.1773 data_time: 0.0099 memory:
4819 loss: 0.5606 decode.loss_ce: 0.3826 decode.acc_seg: 72.6566 aux.loss_ce: 0.1780 aux.acc_seg: 72.6062
```